BIKE SHARING DEMAND ANALYSIS

Submitted By:
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Internship Organization:
NextHikes IT Solutions

Project Type:
Individual Project

Submission Date:
25 March 2026

INDEX PAGE

S. No.	Content	Page No.
1	Certificate	1
2	Declaration	2
3	Acknowledgement	3
4	Introduction	4
5	Objective	5
6	Dataset Description	6
7	Excel Foundations & Data Handling	9-10
8	Data Loading & Data Merging	11-12
9	Data Cleaning	13-14
10	Conditional Logic & Lookup Functions	15-16
11	Advanced Data Analysis	17-18
12	Data Visualization	19-24
13	Dashboard	25-26
14	Final Analysis & Insights	27
15	Conclusion	28
16	Recommendations	29
17	Bibliography	30

DECLARATION

I hereby declare that the project titled *"Bike Sharing Demand Analysis"* has been completed by me as part of my internship at **NextHikes IT Solutions**. This project is my original work and has not been submitted elsewhere for any academic or professional purpose. I have ensured that all the information provided in this report is true and accurate to the best of my knowledge.

Candidate Name: Harshita Gautam

Date: 25 March 2026

CERTIFICATE

This is to certify that **Harshita Gautam** has successfully completed the project titled *"Bike Sharing Demand Analysis"* as part of her internship at NextHikes IT Solutions.

This project is an individual work carried out under guidance and has been completed satisfactorily. It reflects her understanding, dedication, and effort in the field of

data analysis using Microsoft Excel. We appreciate her sincere efforts and wish her success in future endeavors.

Signature Of Internship Mentor: ……

Name of Internship Mentor: Ms/Ms Twinkle

Date: 25 March 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to NextHikes IT Solutions for providing me with the opportunity to work on this project. This experience has been a valuable journey of learning and growth, and I am thankful for the support and guidance received throughout the completion of this project.

I would like to extend my heartfelt thanks to my mentors at NextHikes IT Solutions for their continuous guidance, valuable suggestions, and encouragement. Their support significantly enhanced my understanding and contributed to the successful execution of this project.

I also express my gratitude to the organization for providing the necessary resources, tools, and a professional environment conducive to learning and development. The opportunities and exposure offered played a crucial role in the completion of this project.

This being an individual project, it reflects my dedicated effort, learning, and commitment. I am thankful to everyone who directly or indirectly supported me throughout this journey.

Candidate Name: Harshita Gautam

Date: 25 March 2026

INTRODUCTION

With the increasing demand for efficient and sustainable transportation systems, bike sharing has emerged as a popular solution in urban areas. It provides an eco-friendly, cost-effective, and convenient mode of travel for short distances. Understanding the factors that influence bike rental demand is essential for improving service quality and business planning.

This project, *"Bike Sharing Demand Analysis"*, focuses on analyzing bike rental data to understand how different factors such as weather conditions, time, and holidays impact the demand for bike rentals. By studying these patterns, it becomes easier to make informed decisions and improve operational efficiency.

The project involves working with multiple datasets containing information about bike rentals, weather conditions, and user behaviour. These datasets are cleaned, combined, and analyzed using Microsoft Excel. Various Excel tools and techniques such as data cleaning, pivot tables, charts, and dashboards are used to extract meaningful insights.

As part of my internship at **NextHikes IT Solutions**, this individual project has helped me apply my knowledge of Excel in a practical scenario. It has enhanced my understanding of data handling, analysis, and visualization, enabling me to derive useful insights that can support better decision-making in bike sharing systems.

Figure 4.1: Conceptual representation of bike sharing demand analysis

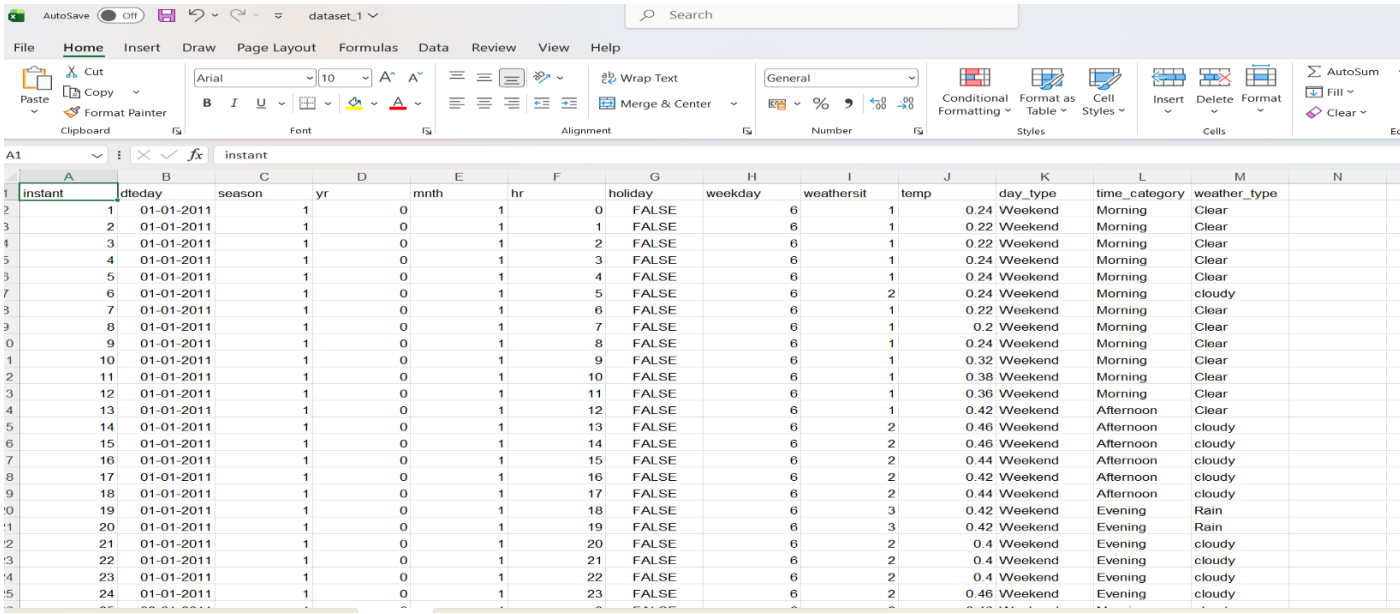
OBJECTIVE

The main objective of this project is to analyze bike sharing data to understand the impact of various factors such as weather conditions, time, and holidays on bike rental demand.

The specific objectives of this project are:

- To understand and apply Excel fundamentals for data handling, including sorting, filtering, and formatting
- To import, merge, and clean multiple datasets using appropriate techniques
- To handle missing and inconsistent data, remove duplicates, and ensure data accuracy
- To apply conditional logic and lookup functions such as IF, VLOOKUP, INDEX, and MATCH
- To perform advanced data analysis using pivot tables, pivot charts, and statistical functions
- To visualize data using charts and create an interactive dashboard for better insights
- To integrate all steps into a final workflow for generating meaningful insights and reports

DATASET DESCRIPTION



instant	dteday	season	yr	mnth	hr	holiday	weekday	weathersit	temp	day_type	time_category	weather_type
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2	01-01-2011	1	0	1	1	FALSE	6	1	0.22	Weekend	Morning	Clear
3	01-01-2011	1	0	1	2	FALSE	6	1	0.22	Weekend	Morning	Clear
4	01-01-2011	1	0	1	3	FALSE	6	1	0.24	Weekend	Morning	Clear
5	01-01-2011	1	0	1	4	FALSE	6	1	0.24	Weekend	Morning	Clear
6	01-01-2011	1	0	1	5	FALSE	6	2	0.24	Weekend	Morning	cloudy
7	01-01-2011	1	0	1	6	FALSE	6	1	0.22	Weekend	Morning	Clear
8	01-01-2011	1	0	1	7	FALSE	6	1	0.2	Weekend	Morning	Clear
9	01-01-2011	1	0	1	8	FALSE	6	1	0.24	Weekend	Morning	Clear
10	01-01-2011	1	0	1	9	FALSE	6	1	0.32	Weekend	Morning	Clear
11	01-01-2011	1	0	1	10	FALSE	6	1	0.38	Weekend	Morning	Clear
12	01-01-2011	1	0	1	11	FALSE	6	1	0.36	Weekend	Morning	Clear
13	01-01-2011	1	0	1	12	FALSE	6	1	0.42	Weekend	Afternoon	Clear
14	01-01-2011	1	0	1	13	FALSE	6	2	0.46	Weekend	Afternoon	cloudy
15	01-01-2011	1	0	1	14	FALSE	6	2	0.46	Weekend	Afternoon	cloudy
16	01-01-2011	1	0	1	15	FALSE	6	2	0.44	Weekend	Afternoon	cloudy
17	01-01-2011	1	0	1	16	FALSE	6	2	0.42	Weekend	Afternoon	cloudy
18	01-01-2011	1	0	1	17	FALSE	6	2	0.44	Weekend	Afternoon	cloudy
19	01-01-2011	1	0	1	18	FALSE	6	3	0.42	Weekend	Evening	Rain
20	01-01-2011	1	0	1	19	FALSE	6	3	0.42	Weekend	Evening	Rain
21	01-01-2011	1	0	1	20	FALSE	6	2	0.4	Weekend	Evening	cloudy
22	01-01-2011	1	0	1	21	FALSE	6	2	0.4	Weekend	Evening	cloudy
23	01-01-2011	1	0	1	22	FALSE	6	2	0.4	Weekend	Evening	cloudy
24	01-01-2011	1	0	1	23	FALSE	6	2	0.46	Weekend	Evening	cloudy

Figure 6: Bike rental dataset showing variables such as date, hour, weather conditions, and rental details in Microsoft Excel

The dataset used in this project is based on bike sharing demand and was provided as part of the internship by **NextHikes IT Solutions**. It consists of multiple datasets containing detailed information about bike rentals, weather conditions, and user behavior over a specific period of time.

The data is recorded on both an hourly and daily basis, which allows for a comprehensive analysis of patterns and trends. This time-based structure helps in understanding how bike rental demand changes throughout the day, across different days of the week, and under varying environmental conditions.

The dataset includes several important variables such as date, hour, season, weather condition, temperature, humidity, wind speed, and total bike rental count. In addition to this, it also provides a breakdown of users into categories such as casual and registered users. This classification helps in analyzing user behavior and identifying differences in usage patterns between different user groups.

Initially, the data was provided in three separate datasets. These datasets were integrated into a single dataset using common keys such as date and time. This merging process enabled a unified and structured dataset for further analysis.

Before performing the analysis, data cleaning techniques were applied to ensure data quality and accuracy. Missing values were identified and handled appropriately, duplicate records were removed, and data types were standardized to maintain consistency. Unnecessary or irrelevant columns were also eliminated to improve the efficiency of the analysis.

The cleaned and combined dataset provided a reliable foundation for performing data analysis and visualization. It enabled the identification of key trends and relationships, such as the impact of weather conditions, time of day, and user types on bike rental demand.

Overall, the dataset plays a crucial role in this project, as it supports meaningful insights and helps in understanding the factors that influence bike sharing usage patterns.

7. EXCEL FOUNDATIONS & DATA HANDLING

Microsoft Excel is a widely used tool for data organization, analysis, and visualization. In this project, Excel played a crucial role in handling and processing the bike rental dataset efficiently.

The dataset was structured in a tabular format consisting of rows and columns, where each row represented an individual record and each column represented a specific variable such as date, season, temperature, and weather conditions. This structure made it easier to manage and analyze the data systematically.

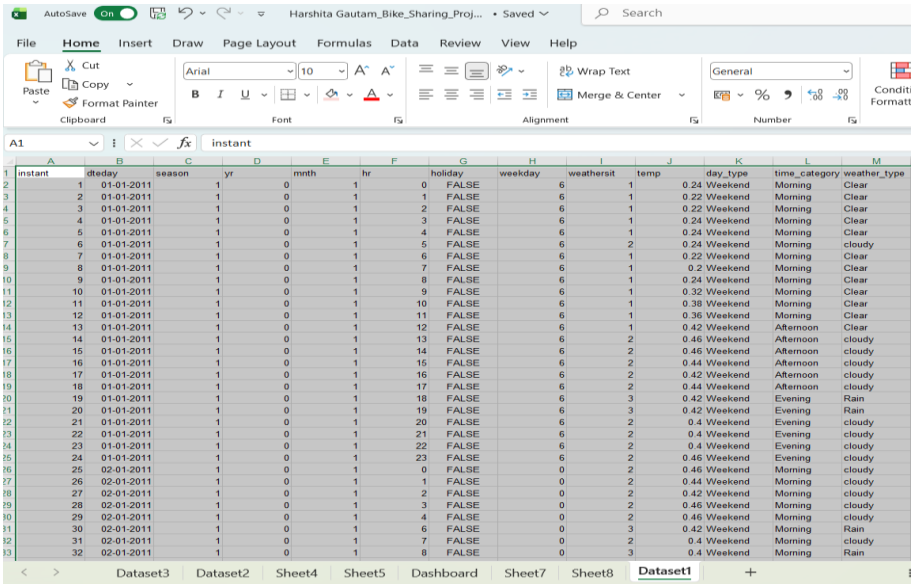
Various data handling techniques were applied to ensure data quality and usability. Sorting and filtering functions were used to organize the dataset and focus on relevant information. These techniques helped in identifying patterns and trends in bike rentals across different conditions.

Data cleaning was an essential part of the process. It involved checking for missing values, removing inconsistencies, and ensuring uniform formatting throughout the dataset. This step improved the accuracy and reliability of the analysis.

Additionally, new columns such as *time_category* and *weather_type* were created using logical conditions to simplify the data and make it more meaningful for analysis. These transformations helped in grouping the data effectively.

Overall, Excel provided a strong foundation for data handling, enabling efficient preparation of the dataset for further analysis, visualization, and dashboard creation.

8. Data Loading



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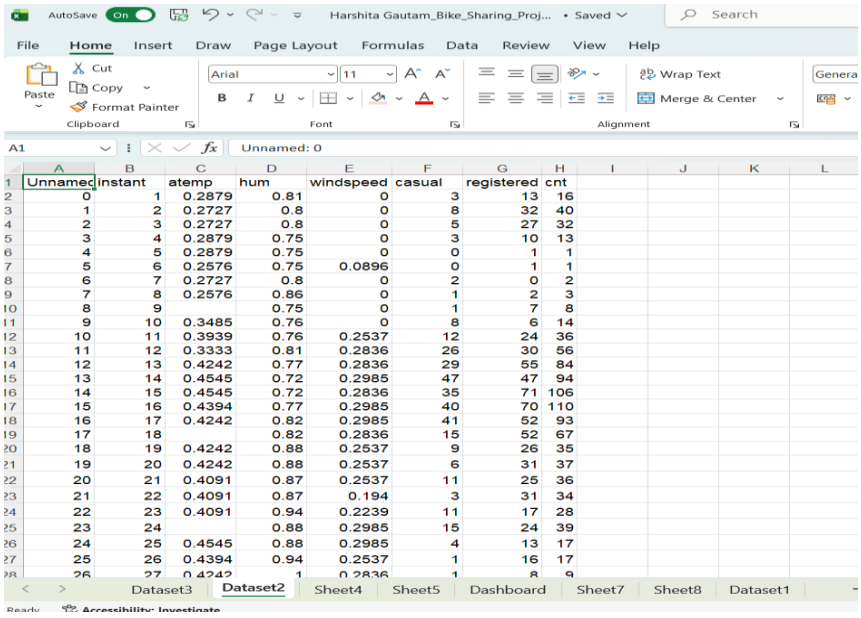
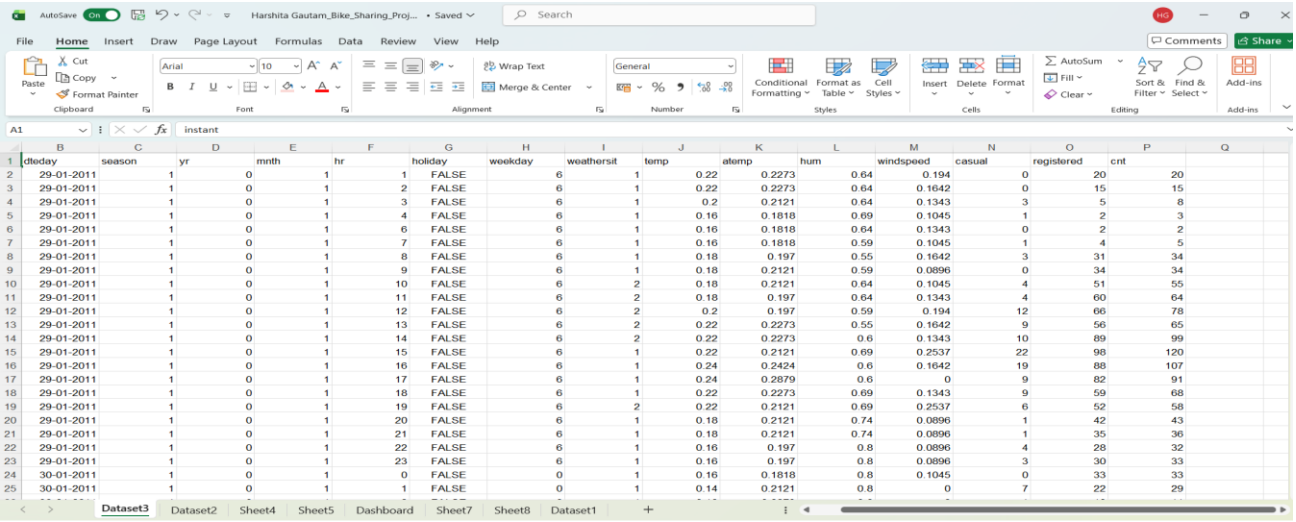
2	01-01-2011	1	0	1	1	FALSE	6	1	0.22	Weekend	Morning	Clear
3	01-01-2011	1	0	1	2	FALSE	6	1	0.22	Weekend	Morning	Clear
4	01-01-2011	1	0	1	3	FALSE	6	1	0.24	Weekend	Morning	Clear
5	01-01-2011	1	0	1	4	FALSE	6	1	0.24	Weekend	Morning	Clear
6	01-01-2011	1	0	1	5	FALSE	6	2	0.24	Weekend	Morning	cloudy
7	01-01-2011	1	0	1	6	FALSE	6	1	0.22	Weekend	Morning	Clear
8	01-01-2011	1	0	1	7	FALSE	6	1	0.2	Weekend	Morning	Clear
9	01-01-2011	1	0	1	8	FALSE	6	1	0.24	Weekend	Morning	Clear
10	01-01-2011	1	0	1	9	FALSE	6	1	0.32	Weekend	Morning	Clear
11	01-01-2011	1	0	1	10	FALSE	6	1	0.38	Weekend	Morning	Clear
12	01-01-2011	1	0	1	11	FALSE	6	1	0.36	Weekend	Morning	Clear
13	01-01-2011	1	0	1	12	FALSE	6	1	0.42	Weekend	Afternoon	Clear
14	01-01-2011	1	0	1	13	FALSE	6	2	0.46	Weekend	Afternoon	cloudy
15	01-01-2011	1	0	1	14	FALSE	6	2	0.46	Weekend	Afternoon	cloudy
16	01-01-2011	1	0	1	15	FALSE	6	2	0.44	Weekend	Afternoon	cloudy
17	01-01-2011	1	0	1	16	FALSE	6	2	0.42	Weekend	Afternoon	cloudy
18	01-01-2011	1	0	1	17	FALSE	6	2	0.44	Weekend	Afternoon	cloudy
19	01-01-2011	1	0	1	18	FALSE	6	3	0.42	Weekend	Evening	Rain
20	01-01-2011	1	0	1	19	FALSE	6	3	0.42	Weekend	Evening	Rain
21	01-01-2011	1	0	1	20	FALSE	6	2	0.4	Weekend	Evening	cloudy
22	01-01-2011	1	0	1	21	FALSE	6	2	0.4	Weekend	Evening	cloudy
23	01-01-2011	1	0	1	22	FALSE	6	2	0.4	Weekend	Evening	cloudy
24	01-01-2011	1	0	1	23	FALSE	6	2	0.46	Weekend	Evening	cloudy
25	02-01-2011	1	0	1	0	FALSE	0	2	0.46	Weekend	Morning	cloudy
26	02-01-2011	1	0	1	1	FALSE	0	2	0.44	Weekend	Morning	cloudy
27	02-01-2011	1	0	1	2	FALSE	0	2	0.42	Weekend	Morning	cloudy
28	02-01-2011	1	0	1	3	FALSE	0	2	0.46	Weekend	Morning	cloudy
29	02-01-2011	1	0	1	4	FALSE	0	2	0.46	Weekend	Morning	cloudy
30	02-01-2011	1	0	1	6	FALSE	0	3	0.42	Weekend	Morning	Rain
31	02-01-2011	1	0	1	7	FALSE	0	2	0.4	Weekend	Morning	cloudy
32	02-01-2011	1	0	1	8	FALSE	0	3	0.4	Weekend	Morning	Rain



Unnamed	instant	atemp	hum	windspeed	casual	registered	cnt
0	1	0.2879	0.81	0	3	13	16
1	2	0.2727	0.8	0	8	32	40
2	3	0.2727	0.8	0	5	27	32
3	4	0.2879	0.75	0	3	10	13
4	5	0.2879	0.75	0	0	1	1
5	6	0.2576	0.75	0.0896	0	1	1
6	7	0.2727	0.8	0	2	0	2
7	8	0.2576	0.86	0	1	2	3
8	9		0.75	0	1	7	8
9	10	0.3485	0.76	0	8	6	14
10	11	0.3939	0.76	0.2537	12	24	36
11	12	0.3333	0.81	0.2836	26	30	56
12	13	0.4242	0.77	0.2836	29	55	84
13	14	0.4545	0.72	0.2985	47	47	94
14	15	0.4545	0.72	0.2836	35	71	106
15	16	0.4394	0.77	0.2985	40	70	110
16	17	0.4242	0.82	0.2985	41	52	93
17	18		0.82	0.2836	15	52	67
18	19	0.4242	0.88	0.2537	9	26	35
19	20	0.4242	0.88	0.2537	6	31	37
20	21	0.4091	0.87	0.2537	11	25	36
21	22	0.4091	0.87	0.194	3	31	34
22	23	0.4091	0.94	0.2239	11	17	28
23	24		0.88	0.2985	15	24	39
24	25	0.4545	0.88	0.2985	4	13	17
25	26	0.4394	0.94	0.2537	1	16	17
26	27	0.4242	1	0.2836	1	8	9



dteday	season	yr	mnth	hr	holiday	weekday	weathersit	temp	atemp	hum	windspeed	casual	registered	cnt
29-01-2011	1	0	1	1	FALSE	6	1	0.22	0.2273	0.64	0.194	0	20	20
29-01-2011	1	0	1	2	FALSE	6	1	0.22	0.2273	0.64	0.1642	0	15	15
29-01-2011	1	0	1	3	FALSE	6	1	0.2	0.2121	0.64	0.1343	3	5	8
29-01-2011	1	0	1	4	FALSE	6	1	0.16	0.1818	0.69	0.1045	1	2	3
29-01-2011	1	0	1	6	FALSE	6	1	0.16	0.1818	0.64	0.1343	0	2	2
29-01-2011	1	0	1	7	FALSE	6	1	0.16	0.1818	0.59	0.1045	1	4	5
29-01-2011	1	0	1	8	FALSE	6	1	0.18	0.197	0.55	0.1642	3	31	34
29-01-2011	1	0	1	9	FALSE	6	1	0.18	0.2121	0.59	0.0896	0	34	34
29-01-2011	1	0	1	10	FALSE	6	2	0.18	0.2121	0.64	0.1045	4	51	55
29-01-2011	1	0	1	11	FALSE	6	2	0.18	0.197	0.64	0.1343	4	60	64
29-01-2011	1	0	1	12	FALSE	6	2	0.2	0.197	0.59	0.194	12	66	78
29-01-2011	1	0	1	13	FALSE	6	2	0.22	0.2273	0.55	0.1642	9	56	65
29-01-2011	1	0	1	14	FALSE	6	2	0.22	0.2273	0.6	0.1343	10	89	99
29-01-2011	1	0	1	15	FALSE	6	1	0.22	0.2121	0.69	0.2537	22	98	120
29-01-2011	1	0	1	16	FALSE	6	1	0.24	0.2424	0.6	0.1642	19	88	107
29-01-2011	1	0	1	17	FALSE	6	1	0.24	0.2879	0.6	0	9	82	91
29-01-2011	1	0	1	18	FALSE	6	1	0.22	0.2273	0.69	0.1343	9	59	68
29-01-2011	1	0	1	19	FALSE	6	2	0.22	0.2121	0.69	0.2537	6	52	58
29-01-2011	1	0	1	20	FALSE	6	1	0.18	0.2121	0.74	0.0896	1	42	43
29-01-2011	1	0	1	21	FALSE	6	1	0.18	0.2121	0.74	0.0896	1	35	36
29-01-2011	1	0	1	22	FALSE	6	1	0.16	0.197	0.8	0.0896	4	28	32
29-01-2011	1	0	1	23	FALSE	6	1	0.16	0.197	0.8	0.0896	3	30	33
30-01-2011	1	0	1	0	FALSE	0	1	0.16	0.1818	0.8	0.1045	0	33	33
30-01-2011	1	0	1	1	FALSE	0	1	0.14	0.2121	0.8	0	7	22	29

Figure 8.1: Dataset loaded into Microsoft Excel for analysis

The first step in this project involved loading the dataset into Microsoft Excel. Multiple datasets were provided and were imported into separate worksheets to maintain proper organization. This helped in managing the data efficiently and allowed easy access during the analysis process.

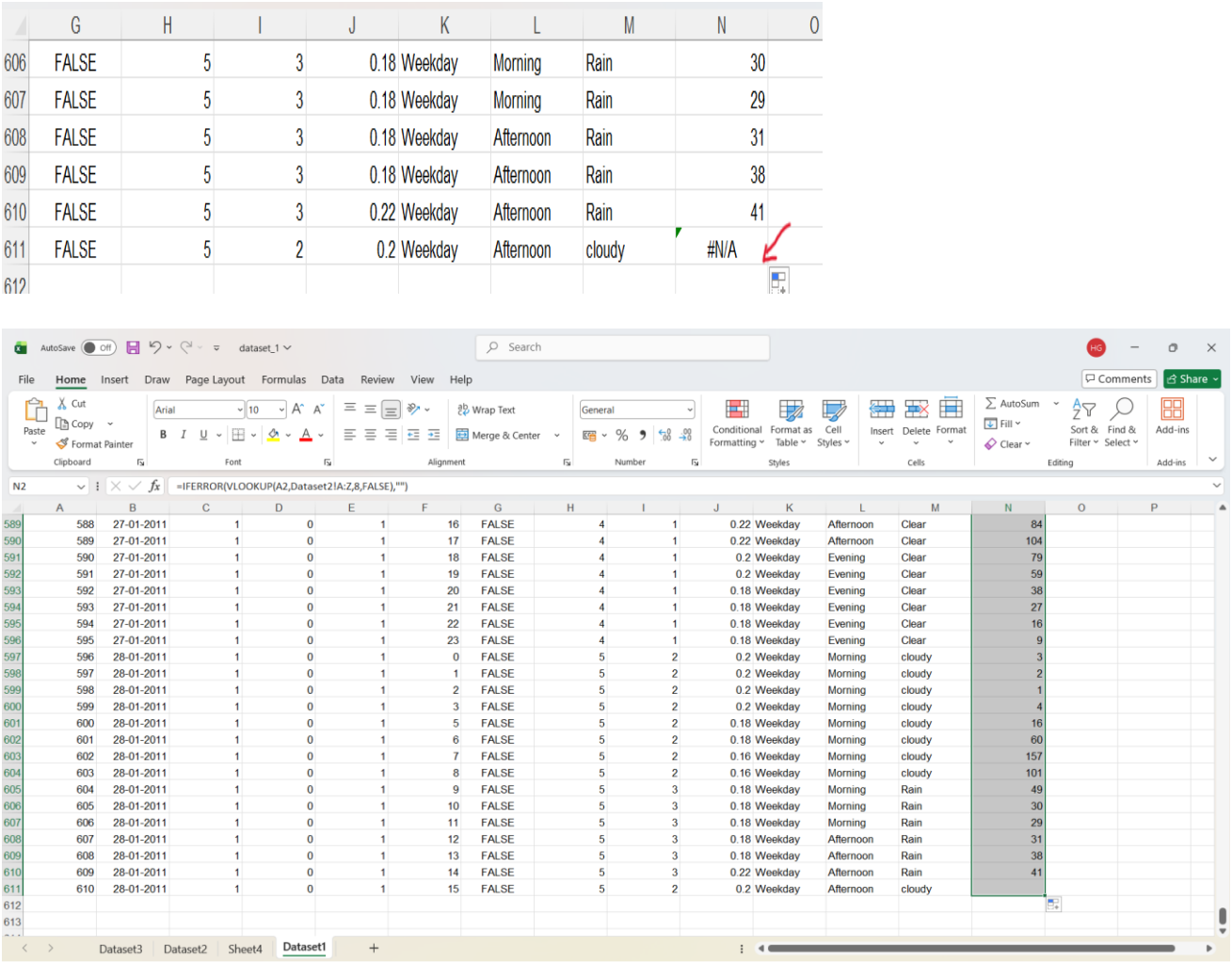
The datasets were structured in tabular form, making it convenient to view and understand different variables such as date, season, temperature, and weather conditions. Proper arrangement of the data ensured a smooth workflow for further processing and analysis.

After loading the data, the next step was to understand the dataset and its columns. The dataset mainly contains information related to bike rentals along with time and weather details. Some of the important columns include:

- **dteday** – shows the date
- **hr** – represents the hour of the day
- **temp and weather** – provide information about weather conditions
- **cnt** – shows the total number of bike rentals

Overall, the dataset shows how bike rental demand changes based on time and weather conditions. Understanding these columns was important before starting the analysis, as it helped in identifying the key factors affecting bike rentals.

8.2 Data Merging



	G	H	I	J	K	L	M	N
606	FALSE	5	3	0.18	Weekday	Morning	Rain	30
607	FALSE	5	3	0.18	Weekday	Morning	Rain	29
608	FALSE	5	3	0.18	Weekday	Afternoon	Rain	31
609	FALSE	5	3	0.18	Weekday	Afternoon	Rain	38
610	FALSE	5	3	0.22	Weekday	Afternoon	Rain	41
611	FALSE	5	2	0.2	Weekday	Afternoon	cloudy	#N/A
612								

Figure 8.2: Data merging performed using lookup functions in Excel

After loading the datasets, the next step was data merging. Since the data was available in multiple sheets, it was necessary to combine relevant information into a single dataset for effective analysis.

Common fields such as unique identifiers (e.g., instant or date) were used to merge the datasets. Functions like VLOOKUP were applied to retrieve important columns, such as total rental counts (cnt), from one dataset to another. This process ensured that all required data was available in one place, making further analysis easier and more efficient.

9. Data Cleaning

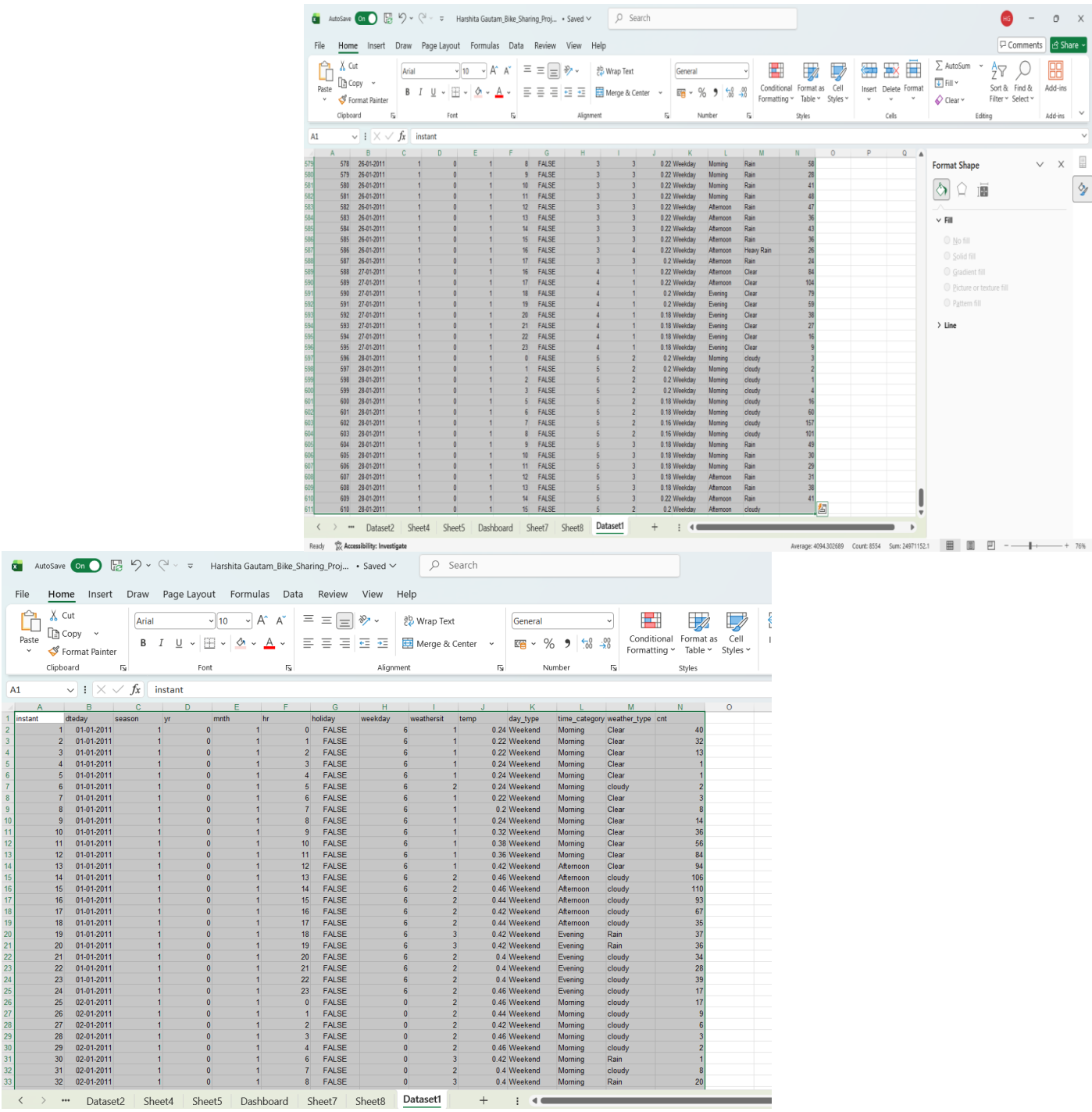


Figure 9.1: Data cleaning process including removal of errors and handling missing values

Data cleaning was performed to ensure the accuracy and consistency of the dataset. The data was carefully checked for missing values, duplicate entries, and any inconsistencies. Errors were corrected to improve the reliability of the dataset.

Additionally, data formatting was applied to maintain uniformity, such as correcting data types (dates and numerical values) and renaming columns for better clarity. New variables like *time_category*, *day_type*, and *weather_type* were also created to simplify the dataset and enhance analysis.

Overall, data cleaning improved the quality of the dataset and ensured that it was ready for further analysis and visualization.

10. Conditional Logic

Conditional logic in Excel is used to perform different actions based on specific conditions. In this project, conditional functions were used to categorize and simplify the dataset for better analysis.

The IF function was mainly used to create new variables such as *time_category* and *day_type*. For example, hours of the day were grouped into categories like Morning, Afternoon, and Evening using logical conditions. Similarly, days were classified as Weekday or Weekend based on their values.

These conditional transformations helped in reducing complexity and made the data easier to analyze. It also allowed grouping of similar data points, which improved the effectiveness of pivot tables and visualizations.

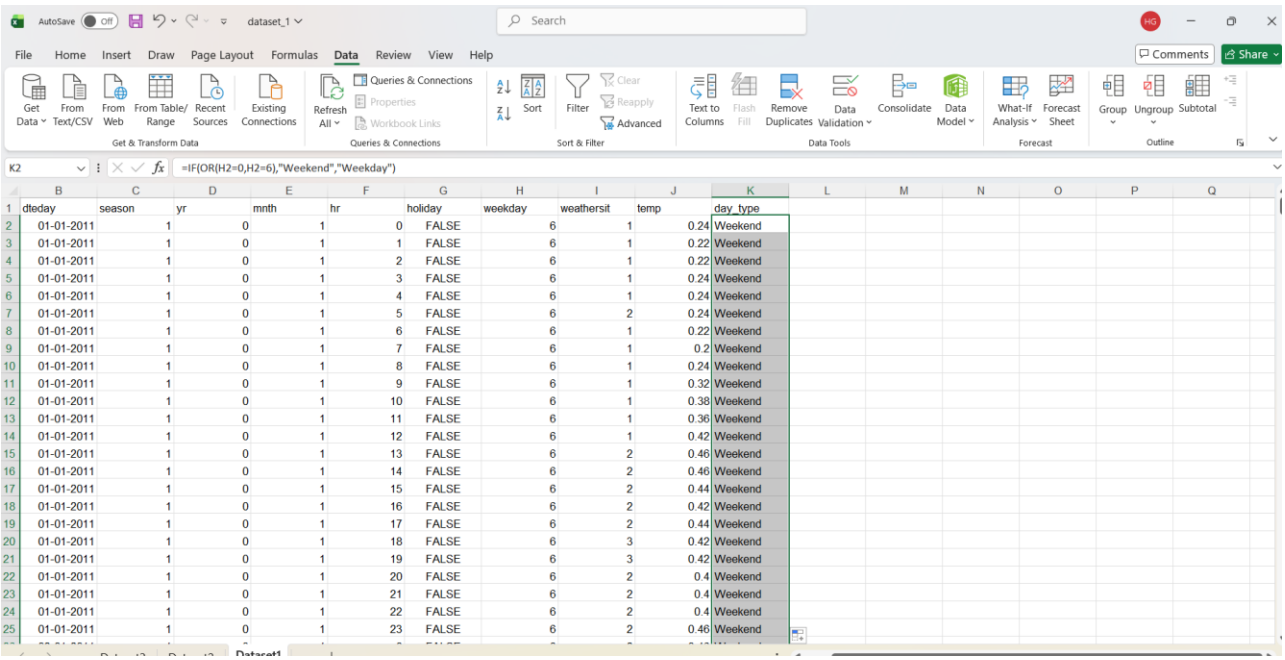


Figure 10.1: Conditional Logic showing use of IF function in Microsoft Excel

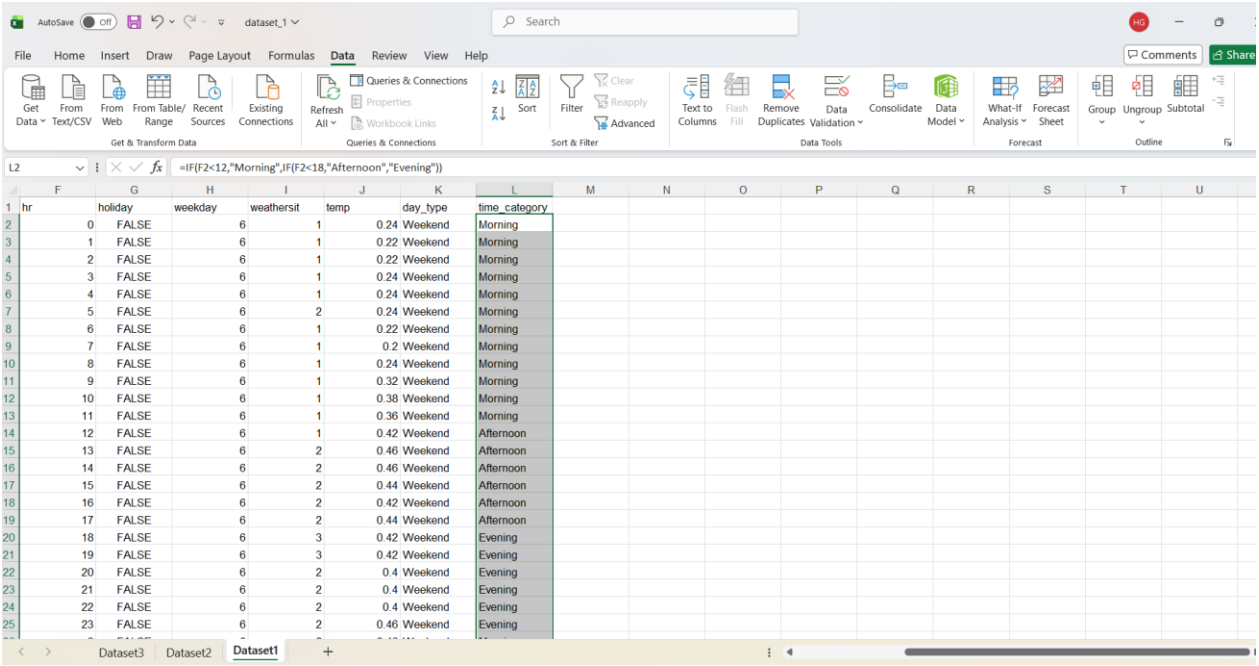
10.2 Lookup Functions

Lookup functions are used to search for specific data in a dataset and retrieve corresponding values. In this project, the VLOOKUP function was used to merge data from multiple datasets.

VLOOKUP helped in retrieving important information such as total rental counts (cnt) from another dataset and adding it to the main dataset. It worked by matching a common field, such as an identifier or date, between two tables.

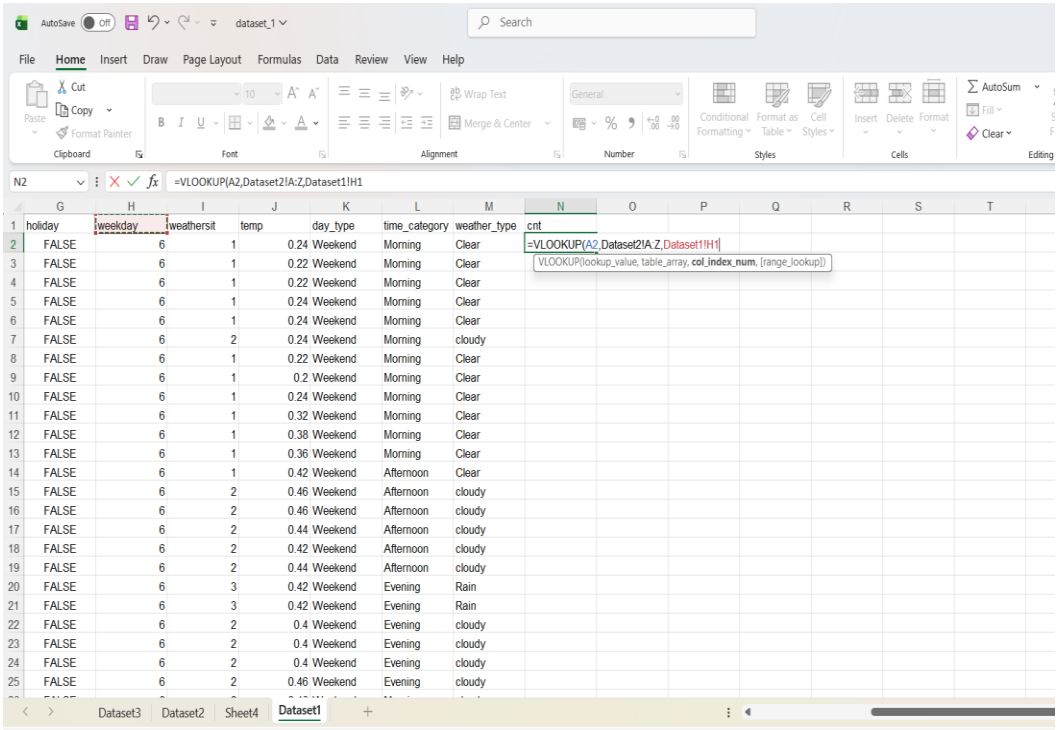
The use of lookup functions reduced manual effort and ensured accuracy in data merging. It also made the dataset more complete and ready for further analysis. Overall, the use of conditional logic and

lookup functions enhanced data organization, improved efficiency, and played a key role in preparing the dataset for analysis and dashboard creation.



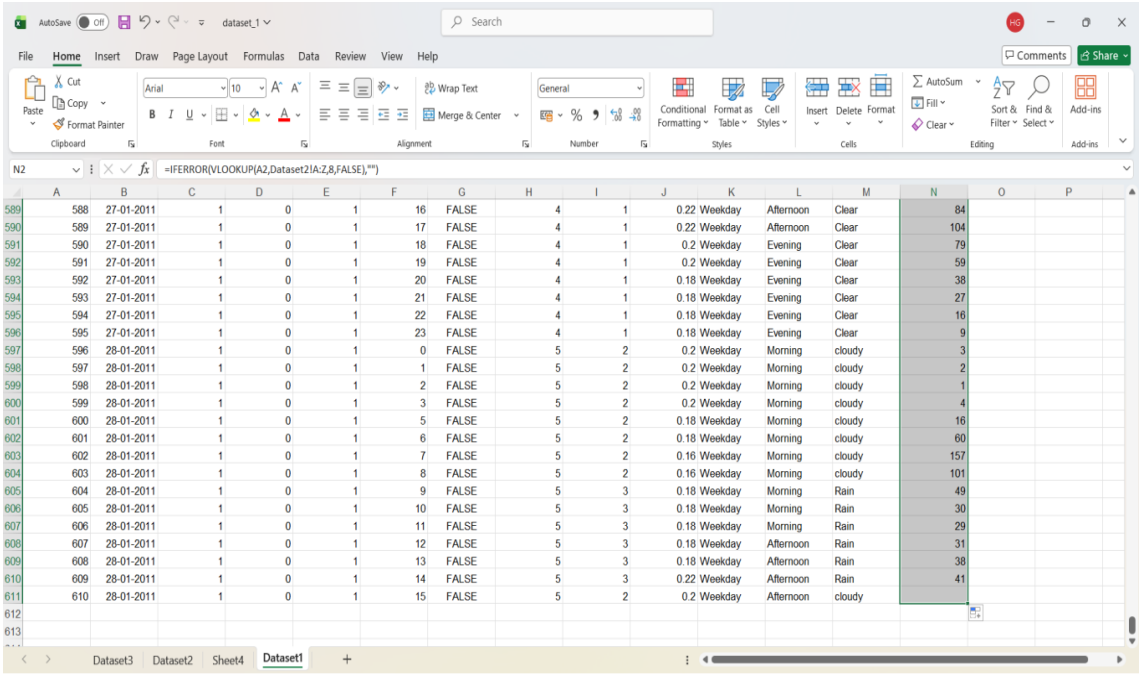
L2 =IF(F2<12,"Morning",IF(F2<18,"Afternoon","Evening"))

hr	holiday	weekday	weathersit	temp	day_type	time_category
0	FALSE	6	1	0.24	Weekend	Morning
1	FALSE	6	1	0.22	Weekend	Morning
2	FALSE	6	1	0.22	Weekend	Morning
3	FALSE	6	1	0.24	Weekend	Morning
4	FALSE	6	1	0.24	Weekend	Morning
5	FALSE	6	2	0.24	Weekend	Morning
6	FALSE	6	1	0.22	Weekend	Morning
7	FALSE	6	1	0.2	Weekend	Morning
8	FALSE	6	1	0.24	Weekend	Morning
9	FALSE	6	1	0.32	Weekend	Morning
10	FALSE	6	1	0.38	Weekend	Morning
11	FALSE	6	1	0.36	Weekend	Morning
12	FALSE	6	1	0.42	Weekend	Afternoon
13	FALSE	6	2	0.46	Weekend	Afternoon
14	FALSE	6	2	0.46	Weekend	Afternoon
15	FALSE	6	2	0.44	Weekend	Afternoon
16	FALSE	6	2	0.42	Weekend	Afternoon
17	FALSE	6	2	0.44	Weekend	Afternoon
18	FALSE	6	3	0.42	Weekend	Evening
19	FALSE	6	3	0.42	Weekend	Evening
20	FALSE	6	2	0.4	Weekend	Evening
21	FALSE	6	2	0.4	Weekend	Evening
22	FALSE	6	2	0.4	Weekend	Evening
23	FALSE	6	2	0.46	Weekend	Evening



N2 =VLOOKUP(A2,Dataset2!A:Z,Dataset1!H1

holiday	weekday	weathersit	temp	day_type	time_category	weather_type	cnt
FALSE	6	1	0.24	Weekend	Morning	Clear	=VLOOKUP(A2,Dataset2!A:Z,Dataset1!H1
FALSE	6	1	0.22	Weekend	Morning	Clear	
FALSE	6	1	0.22	Weekend	Morning	Clear	
FALSE	6	1	0.24	Weekend	Morning	Clear	
FALSE	6	1	0.24	Weekend	Morning	Clear	
FALSE	6	2	0.24	Weekend	Morning	cloudy	
FALSE	6	1	0.22	Weekend	Morning	Clear	
FALSE	6	1	0.2	Weekend	Morning	Clear	
FALSE	6	1	0.24	Weekend	Morning	Clear	
FALSE	6	1	0.32	Weekend	Morning	Clear	
FALSE	6	1	0.38	Weekend	Morning	Clear	
FALSE	6	1	0.36	Weekend	Morning	Clear	
FALSE	6	1	0.42	Weekend	Afternoon	Clear	
FALSE	6	2	0.46	Weekend	Afternoon	cloudy	
FALSE	6	2	0.46	Weekend	Afternoon	cloudy	
FALSE	6	2	0.44	Weekend	Afternoon	cloudy	
FALSE	6	2	0.42	Weekend	Afternoon	cloudy	
FALSE	6	2	0.44	Weekend	Afternoon	cloudy	
FALSE	6	3	0.42	Weekend	Evening	Rain	
FALSE	6	3	0.42	Weekend	Evening	Rain	
FALSE	6	2	0.4	Weekend	Evening	cloudy	
FALSE	6	2	0.4	Weekend	Evening	cloudy	
FALSE	6	2	0.4	Weekend	Evening	cloudy	
FALSE	6	2	0.46	Weekend	Evening	cloudy	

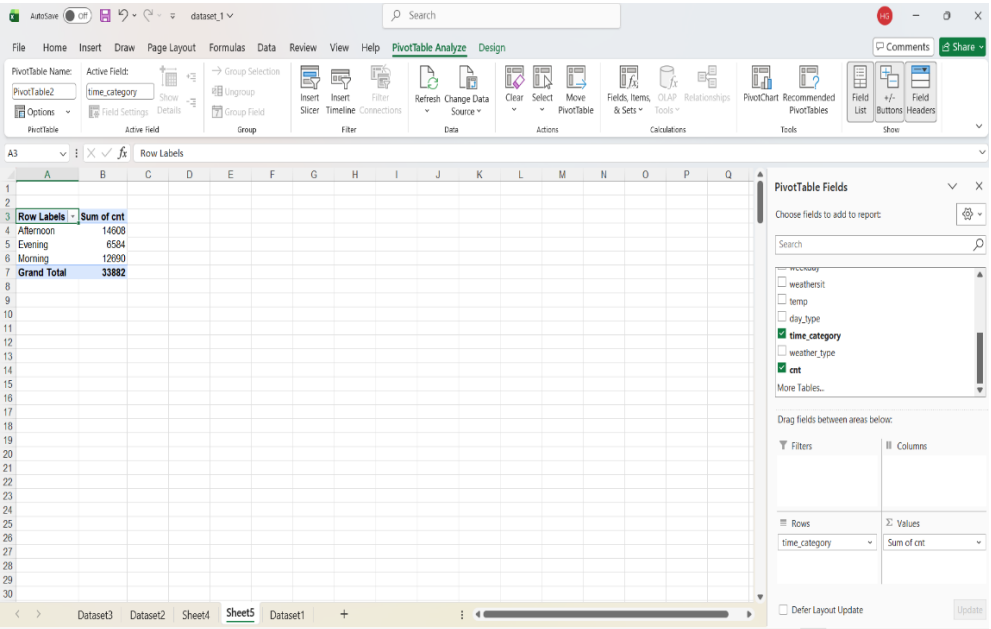


N2 =IFERROR(VLOOKUP(A2,Dataset2!A:Z,8,FALSE),"")

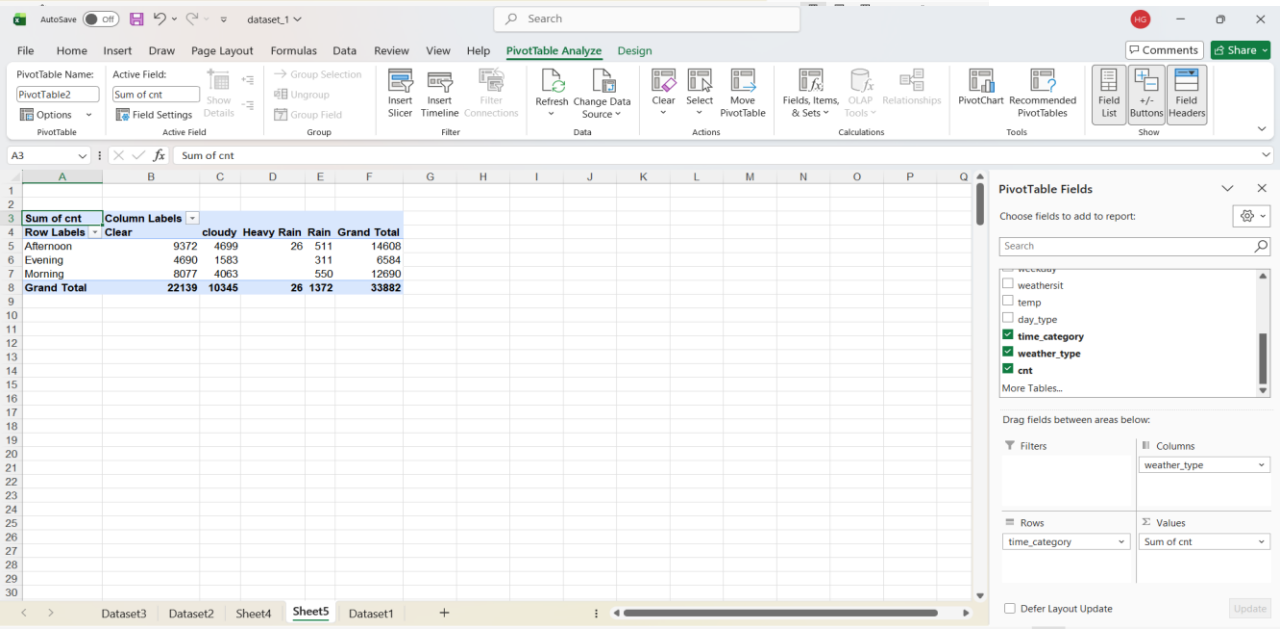
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
589	588	27-01-2011	1	0	1	16	FALSE	4	1	0.22	Weekday	Afternoon	Clear	84
590	589	27-01-2011	1	0	1	17	FALSE	4	1	0.22	Weekday	Afternoon	Clear	104
591	590	27-01-2011	1	0	1	18	FALSE	4	1	0.2	Weekday	Evening	Clear	79
592	591	27-01-2011	1	0	1	19	FALSE	4	1	0.2	Weekday	Evening	Clear	59
593	592	27-01-2011	1	0	1	20	FALSE	4	1	0.18	Weekday	Evening	Clear	38
594	593	27-01-2011	1	0	1	21	FALSE	4	1	0.18	Weekday	Evening	Clear	27
595	594	27-01-2011	1	0	1	22	FALSE	4	1	0.18	Weekday	Evening	Clear	16
596	595	27-01-2011	1	0	1	23	FALSE	4	1	0.18	Weekday	Evening	Clear	9
597	596	28-01-2011	1	0	1	0	FALSE	5	2	0.2	Weekday	Morning	cloudy	3
598	597	28-01-2011	1	0	1	1	FALSE	5	2	0.2	Weekday	Morning	cloudy	2
599	598	28-01-2011	1	0	1	2	FALSE	5	2	0.2	Weekday	Morning	cloudy	1
600	599	28-01-2011	1	0	1	3	FALSE	5	2	0.2	Weekday	Morning	cloudy	4
601	600	28-01-2011	1	0	1	5	FALSE	5	2	0.18	Weekday	Morning	cloudy	16
602	601	28-01-2011	1	0	1	6	FALSE	5	2	0.18	Weekday	Morning	cloudy	60
603	602	28-01-2011	1	0	1	7	FALSE	5	2	0.16	Weekday	Morning	cloudy	157
604	603	28-01-2011	1	0	1	8	FALSE	5	2	0.16	Weekday	Morning	cloudy	101
605	604	28-01-2011	1	0	1	9	FALSE	5	3	0.18	Weekday	Morning	Rain	49
606	605	28-01-2011	1	0	1	10	FALSE	5	3	0.18	Weekday	Morning	Rain	30
607	606	28-01-2011	1	0	1	11	FALSE	5	3	0.18	Weekday	Morning	Rain	29
608	607	28-01-2011	1	0	1	12	FALSE	5	3	0.18	Weekday	Afternoon	Rain	31
609	608	28-01-2011	1	0	1	13	FALSE	5	3	0.18	Weekday	Afternoon	Rain	38
610	609	28-01-2011	1	0	1	14	FALSE	5	3	0.22	Weekday	Afternoon	Rain	41
611	610	28-01-2011	1	0	1	15	FALSE	5	2	0.2	Weekday	Afternoon	cloudy	

Figure 10.2: Lookup Functions showing use of VLOOKUP/XLOOKUP for data retrieval

ADVANCED DATA ANALYSIS



Row Labels	Sum of cnt
Afternoon	14608
Evening	6584
Morning	12690
Grand Total	33882



Sum of cnt	Column Labels				
Row Labels	Clear	cloudy	Heavy Rain	Rain	Grand Total
Afternoon	9372	4699	26	511	14608
Evening	4690	1583		311	6584
Morning	8077	4063		550	12690
Grand Total	22139	10345	26	1372	33882

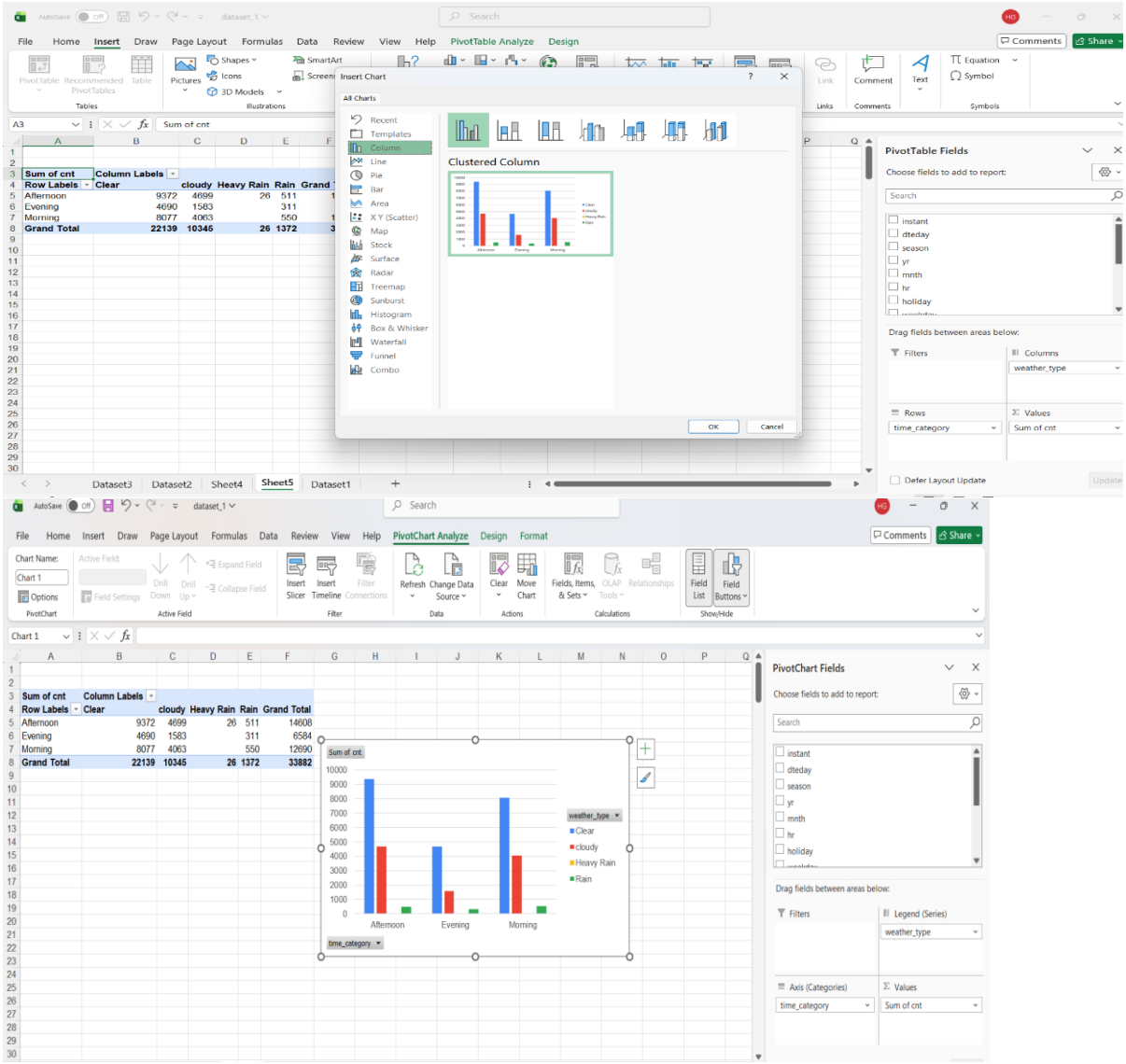


Figure 11.1: Advanced Data Analysis using Pivot Table

Advanced data analysis involves extracting meaningful insights from the dataset using various analytical tools and techniques in Excel. In this project, advanced analysis was performed to understand patterns and trends in bike rentals.

Pivot tables were extensively used to summarize and analyze large amounts of data efficiently. They helped in grouping the data based on different categories such as *time_category*, *weather_type*, and *day_type*. This made it easier to compare bike rental demand across different conditions.

Different calculations such as total rentals and comparisons between categories were performed using pivot tables. This allowed identification of key trends, such as which time of the day had the highest demand and how weather conditions affected bike rentals.

Filtering and sorting features were also used to focus on specific data segments and gain deeper insights. These techniques helped in breaking down the dataset into smaller, more manageable parts for detailed analysis.Overall, advanced data analysis helped in transforming raw data into meaningful information, which was essential for making informed observations and supporting the final insights of the project.

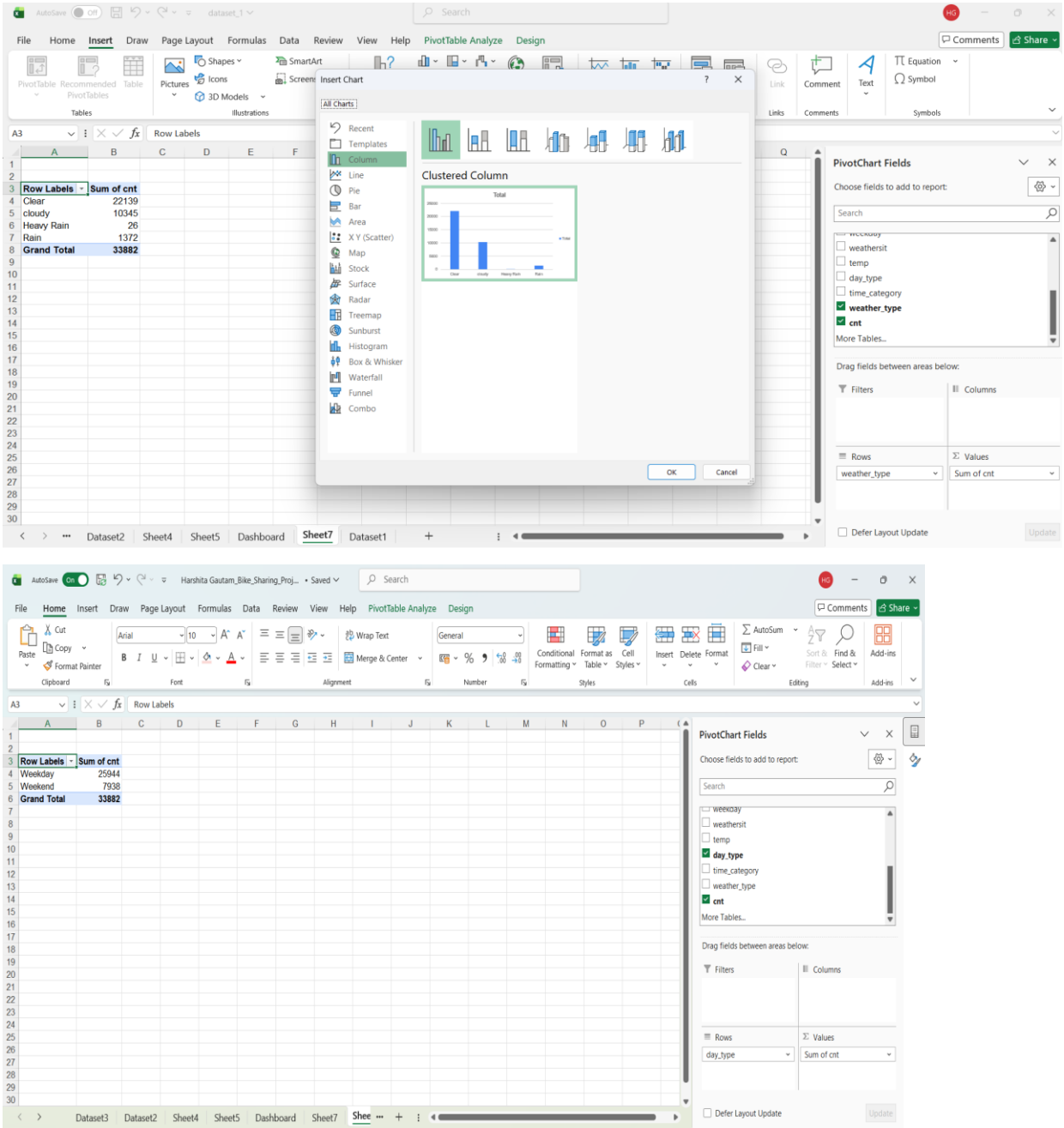
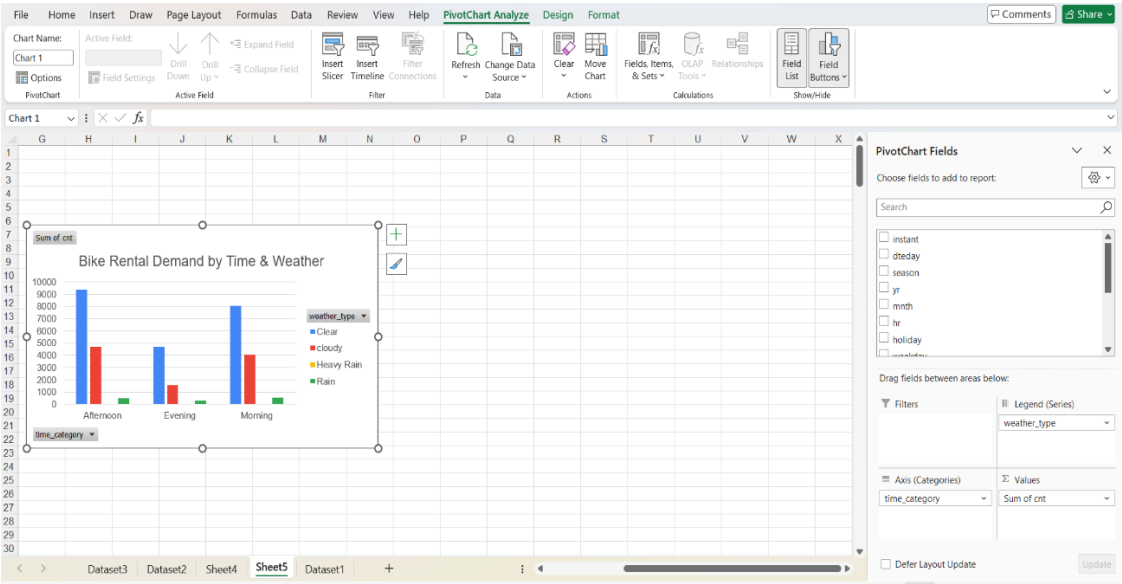
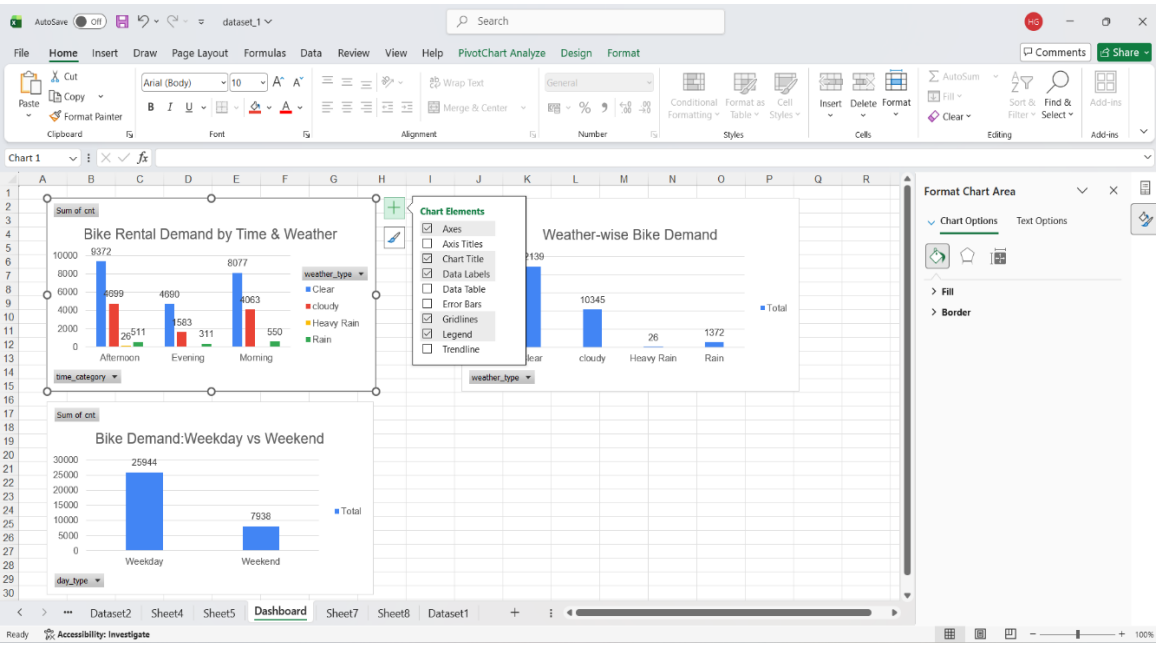


Figure 11.2: Data summarization and grouping for analysis

12. Data Visualization





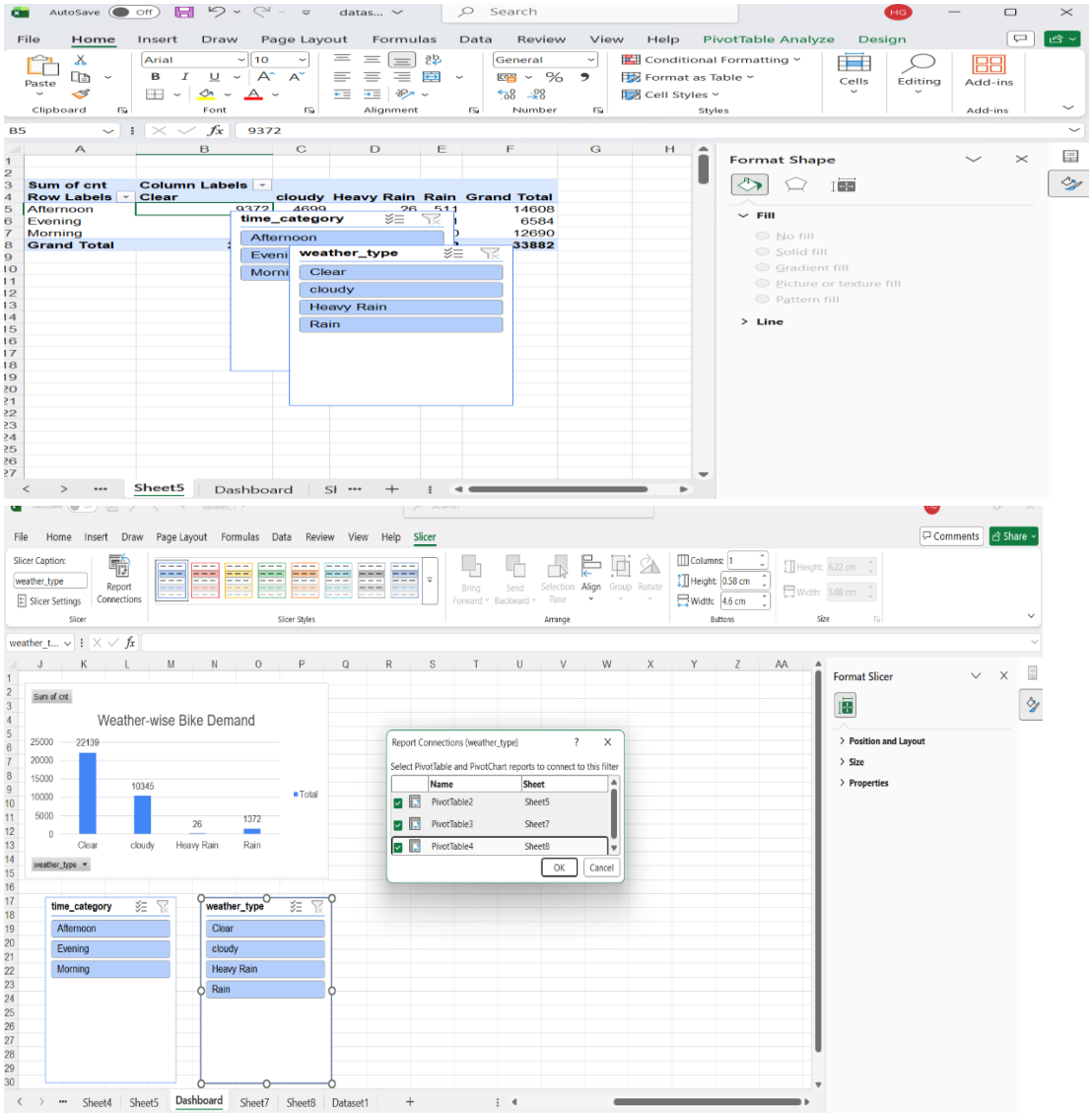


Figure 12.1: Data Visualization showing bike rental trends using charts

Data visualization is an essential part of data analysis as it helps in presenting data in a graphical and easy-to-understand format. In this project, various charts were created in Microsoft Excel to represent the bike rental data visually.

Column charts were used to compare bike rental demand across different categories such as *time_category*, *weather_type*, and *day_type*. These visualizations helped in identifying patterns and trends in the dataset.

The use of charts made complex data more understandable and allowed quick comparison between different variables. It also helped in highlighting key observations effectively.

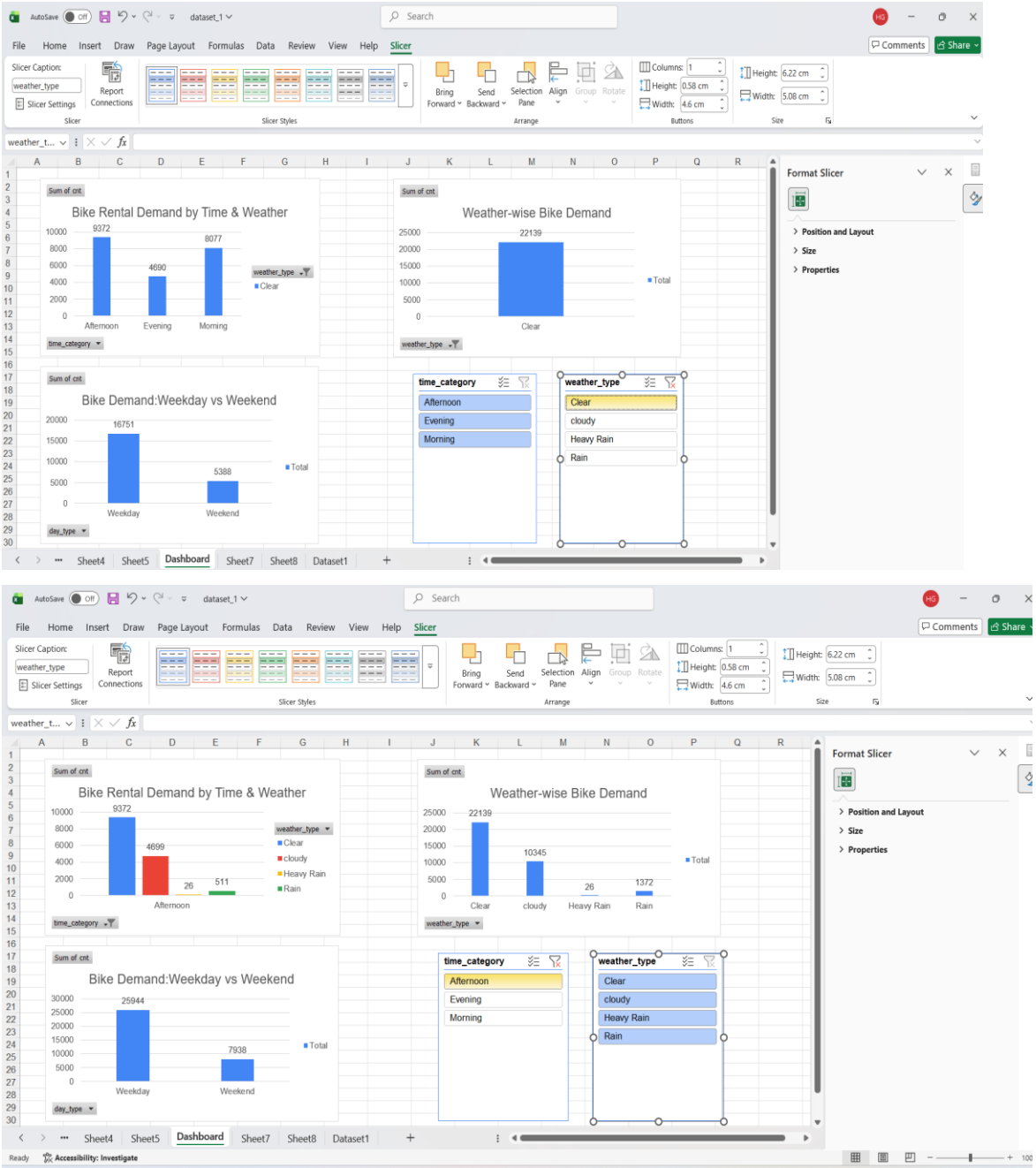
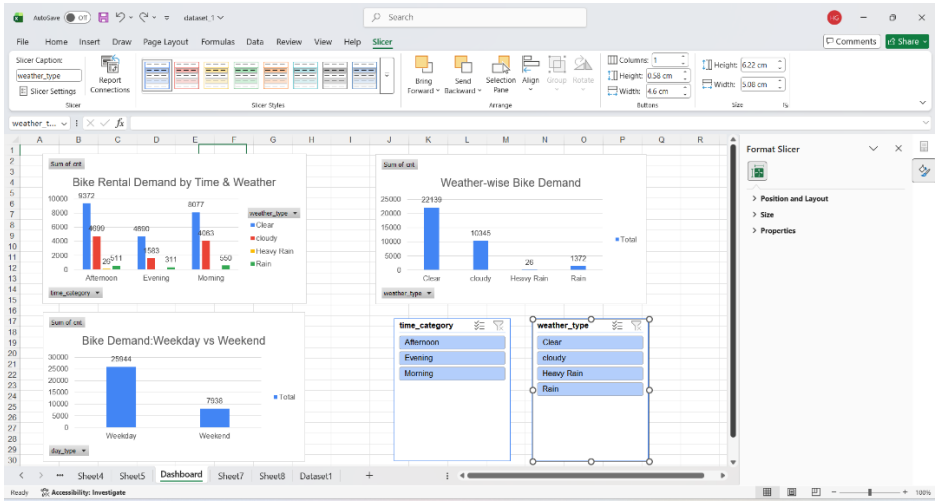


Figure 12.2: Graphical representation of bike rental data

14. Dashboard



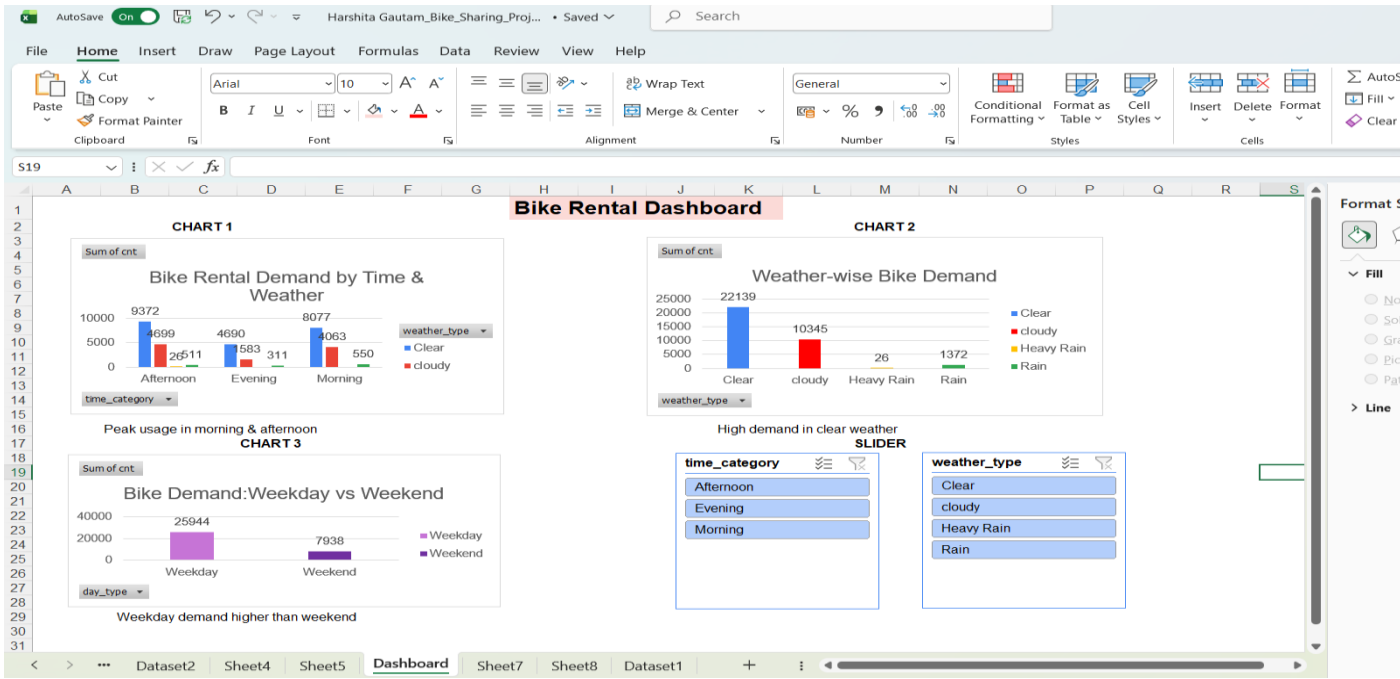


Figure 13: Interactive dashboard with slicers for dynamic data analysis

A dashboard was created to combine all visual elements into a single interactive interface. It provided a summarized view of the analysis and made it easier to interpret the results.

The dashboard included multiple charts displaying bike rental demand based on time of day, weather conditions, and weekday versus weekend comparisons. Slicers were added to enable interactive filtering, allowing users to dynamically explore the data.

Proper formatting, alignment, and color coding were applied to enhance the overall appearance and readability of the dashboard. Consistent color schemes were maintained across all charts for better understanding.

Overall, the dashboard served as an effective tool for presenting insights in a clear, organized, and interactive manner.

14. Final Analysis & Insights

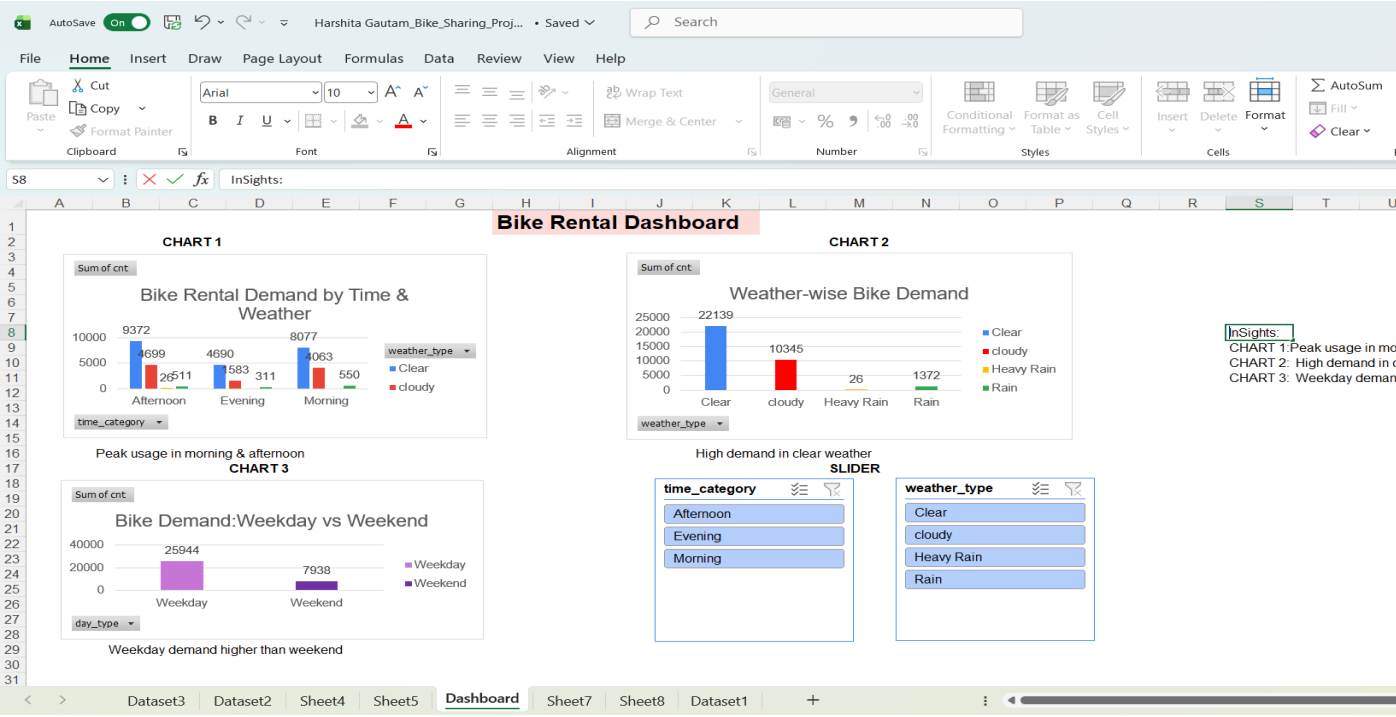


Figure 14: Final Analysis & Insights derived from the dashboard

The analysis of the bike rental dataset provides valuable insights into user behavior and the key factors influencing bike rental demand. By utilizing pivot tables, charts, and an interactive dashboard, meaningful patterns and trends were identified effectively.

One of the major observations from the analysis is that bike rental demand is highest during the **morning and afternoon hours**, indicating peak usage during typical commuting periods. This suggests that a large number of users prefer renting bikes for daily activities such as traveling to work, college, or other routine purposes. In comparison, the demand decreases during the evening, showing a clear variation based on the time of day.

Weather conditions were found to have a significant impact on bike rental usage. The data reveals that the highest number of rentals occurs during **clear weather conditions**, highlighting a strong positive influence on user behavior. On the other hand, demand decreases during **cloudy weather** and drops drastically during **rainy and heavy rain conditions**. This clearly indicates that unfavorable weather discourages users from renting bikes.

Another important insight is observed in the comparison between weekdays and weekends. The analysis shows that bike rental demand is considerably higher on **weekdays** compared to weekends. This pattern suggests that the service is primarily used for **daily commuting rather than recreational purposes**, as users tend to rely less on bike rentals during weekends.

Furthermore, the use of categorized variables such as *time_category, weather_type,* and *day_type* played a crucial role in simplifying the analysis. These categories helped in organizing the data efficiently and enabled better comparison across different conditions, leading to clearer insights.

The interactive dashboard developed in this project enhanced data visualization and interpretation. With the help of slicers, users can dynamically explore different scenarios based on time and weather conditions, making the analysis more flexible and user-friendly.

Conclusion

In conclusion, this project successfully analyzed the bike rental dataset using Microsoft Excel as the primary tool. Various data processing techniques, including data loading, cleaning, transformation, and analysis, were effectively applied to ensure accurate and meaningful results.

The use of pivot tables, charts, and an interactive dashboard enabled clear identification of patterns and trends in bike rental demand. Key factors such as **time of day, weather conditions, and type of day (weekday vs weekend)** were found to significantly influence user behavior and rental patterns.

Additionally, this project highlights the importance of **data visualization** in simplifying complex data and presenting insights in an understandable and impactful manner. The interactive dashboard further enhanced usability by allowing dynamic exploration of the dataset.

Overall, the analysis provided valuable insights into bike rental usage, helping to better understand customer behavior. These findings can support informed decision-making, such as optimizing bike availability during peak hours and planning operations based on weather conditions.

Recommendations

Based on the analysis of the bike rental dataset, the following recommendations are suggested to improve operational efficiency and user satisfaction:

1. Optimize Bike Availability During Peak Hours

- Since demand is highest during **morning and afternoon**, more bikes should be made available during these time periods.
- Proper distribution of bikes can help reduce shortages and improve customer experience.

2. Weather-Based Planning

- As demand is highest in **clear weather** and drops during rainy conditions, companies should plan operations accordingly.
- During unfavorable weather, maintenance activities or reduced fleet deployment can be scheduled.

3. Focus on Weekday Demand

- Higher demand on **weekdays** indicates strong usage for commuting.
- Services can be optimized for office and college routes, ensuring better availability in high-demand areas.

4. Promotional Strategies for Weekends

- Since demand is lower on weekends, special **discounts, offers, or promotional campaigns** can be introduced to attract more users.
- This can help balance demand across the week.

5. Improve Data Monitoring and Dashboard Usage

- Regular monitoring using dashboards can help in **real-time decision-making**.
- Continuous data tracking will allow businesses to quickly adapt to changing trends.

6. Enhance User Experience

- Ensure bikes are well-maintained and easily accessible during peak hours.
- Introduce features like **easy booking, tracking, and flexible pricing** to improve customer satisfaction.

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